Taichi Kamei

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Embedded systems & Hardware engineer who takes projects from 0 to 1 under real physical constraints

Education

4th Year Engineering Physics *Expected Graduation - May 2028*
University of British Columbia | Vancouver

Relevant Courses: Electric Circuit Analysis, Signals and Systems, Software Construction, Digital Systems and Microcomputers, Thermodynamics, Statistical Mechanics, Quantum Mechanics, Applied Linear Algebra

Experience

Firmware & Hardware Engineer Co-op *May 2026 - Present*
UBC Blusson Quantum Matter Institute

- Initially started as a PCB validation role for the 4-channel voltage controlled biasing current source PCB used for the superconducting nanowire single photon detector (SNSPD)
- Recognized the need for firmware development, designed an active object architecture in C/C++ on ESP32 with ESP-IDF, FreeRTOS, dual-core task isolation, inter-core queue communication, and event coordination, achieving deterministic concurrent per-channel control
- Ported Analog Devices no-OS 24-bit Sigma-Delta ADC and 16-bit DAC drivers to ESP-IDF, implemented SPI platform abstraction layer with manual CS toggle for 5 SPI slaves and mixed SPI modes on a 10MHz shared bus
- Diagnosed ADC garbage reading due to SPI CPOL mismatch on different SPI modes (DAC: Mode 1, ADC: Mode 3), resolved the issue by adding an NOP transaction before actual transaction
- Developed hysteresis V-I sweep mode with 10mV step size and steady-state biasing mode with 10nA precision
- Designed calibration mode using Non-Volatile Storage, enabling runtime calibration lasting across power-cycle
- Implemented UART connection watchdog, resetting all DAC to 0.0V when USB cable is unplugged for 2.5 seconds
- Debugged 3.3V LDO undervoltage by oscilloscope scoping and datasheet comparison, applied hardware workaround and PCB layout revision on Altium
- Tested time domain DAC and ADC drift charateristics by logging both PCB and commercial SMU measurement for 2 days using Raspberry Pi

Projects

Drone Flight Controller *Feb 2026 - Present*

- Designed a 4-layer flight controller PCB in KiCAD, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261 chip
- Implemented power regulation stage for 4S LiPo input with 5V/5A buck converter, 3.3V LDO for sensors and MCU, and a split ground for minimizing noise from high current 15V ESC line
- Designed a 4-layer GPS-module & magnetometer PCB fitting within 260mm by 260mm

Autonomous Clue Detecting Robot *November 2025*

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

Autonomous Payload Retrieving Robot *June - August 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

Skills

Software	C/C++, Python, Java, Assembly, VHDL, Linux, Git, CMake
Embedded	ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, ROS, Gazebo, FPGA, I2C, SPI, UART
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, Electrometer, DMM, SMU, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper